Assignment 11

Total = 10 Marks

Multiple Choice Questions

Question 1: Consider the following two statements:

Statement A: SVMs are less prone to be trapped in local minima as compared to MLP.
Statement B: SVMs are less prone to overfitting as compared to MLP.

Which of the following options is correct?

(a) Statement A is correct and Statement B is incorrect
(b) Statement A is incorrect and Statement B is correct
(c) Both the Statements are correct
(d) Both the Statements are incorrect

[2 *Mark*]

Correct answer: (c) Both the Statements are correct

Detailed Solution:

SVM training involves solving a convex quadratic programming problem. This guarantees finding a global minimum, whereas MLPs rely on non-convex optimization (such as gradient descent with backpropagation) and can frequently get trapped in local minima.

SVMs are designed to maximize the margin between classes (or control the epsilon-insensitive error tube in regression), which implements structural risk minimization. MLPs primarily focus on minimizing empirical risk (training error), making them traditionally more prone to overfitting unless carefully regularized.

Question 2: Consider the following two statements:

Statement A: In hard clustering, membership of a data point is binary (0 or 1), whereas in soft clustering, it can belong to multiple clusters with different probabilities.
Statement B: Hierarchical clustering algorithms are highly robust, making them capable of automatically correcting previous misclassifications as the tree structure is built.

Which of the following options is correct?

(a) Statement A is correct and Statement B is incorrect
(b) Statement A is incorrect and Statement B is correct
(c) Both the Statements are correct
(d) Both the Statements are incorrect

[2 *Mark*]

Correct answer: (a) Statement A is correct and Statement B is incorrect

Detailed Solution:

Hard clustering assigns a strict binary membership (0 or 1) to data points, while soft clustering allows an object to belong to multiple clusters with different probabilities.

Classical hierarchical clustering (HC) algorithms, lack robustness and are highly sensitive to noise and outliers. Furthermore, a major limitation of HC algorithms is that they are entirely incapable of correcting previous misclassifications once a data point has been assigned to a cluster.

Question 3: Consider a dataset consisting of three points with standardized features x_1 and x_2 given in table.

Point	x_1	x_2
P_1	1	2
P_2	2	2
P_3	6	5

Apply K-means clustering with $K = 2$ using squared Euclidean distance. In a particular iteration, the centroids of *Cluster* 1 is $C_1 \equiv (1,2)$, and that of *Cluster* 2 is $C_2 \equiv (2,2)$. What are the cluster assignments of each point?

(a) $Cluster\ 1{:}\{P_1, P_2\}\ and\ Cluster\ 2{:}\{P_3\}$
(b) $Cluster\ 1{:}\{P_1, P_3\}\ and\ Cluster\ 2{:}\{P_2\}$
(c) $Cluster\ 1{:}\{P_1\}\ and\ Cluster\ 2{:}\{P_2, P_3\}$
(d) $Cluster\ 1{:}\{P_1, P_2, P_3\}\ and\ Cluster\ 2{:}\{No\ points\}$

[3 *Mark*]

Correct answer: (c) $\boldsymbol{Cluster\ 1{:}\{P_1\}\ and\ Cluster\ 2{:}\{P_2, P_3\}}$

Detailed Solution:

We calculate the distance between each point and the centroids.

For computational efficiency in K-means, the squared Euclidean distance is used:

$$d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2$$

Iteration 1: Using $C_1 = (1,2)$ and $C_2 = (2,2)$

For $P_1(1,2)$:

Distance to C_1: $d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2 = (1-1)^2 + (2-2)^2 = 0$

Distance to C_2: $d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2 = (1-2)^2 + (2-2)^2 = 1$

Since, Distance to C_1 < Distance to C_2, so, the assigned cluster is C_1.

Similarly, for all the other points calculations were carried out and results are tabulated.

Point	$d^2(C_1)$	$d^2(C_2)$	**Cluster Assigned**
P_1	0	1	C_1
P_2	1	0	C_2
P_3	34	25	C_2

Question 4: A satellite image is classified into three land-cover classes ($c = 3$), $C_1 = Water$; $C_2 = Vegetation$; and $C_3 = Built\text{-}up$. Two normalized spectral variables x_1 and x_2 are used. Using the One-Versus-One approach, binary classifiers were trained for every possible pair of classes. The decision boundaries for these classifiers are given below:

Classifier Pair	**Decision function $f_{ij}(x)$**	**Rule (vote)**
C_1 vs C_2	$f_{12}(x_1, x_2) = x_1 - x_2 - 0.5$	If $f_{12} > 0$, vote C_1; else vote C_2
C_1 vs C_3	$f_{13}(x_1, x_2) = x_1 + x_2 - 3$	If $f_{13} > 0$, vote C_1; else vote C_3
C_2 vs C_3	$f_{23}(x_1, x_2) = 2x_2 - x_1 - 1$	If $f_{23} > 0$, vote C_2; else vote C_3

A new unclassified pixel has normalized spectral variables as ($x_1 = 2.5$, $x_2 = 1.2$).

Determine the vote obtained from each classifier and the final class of the new pixel.

(a) $f_{12} \rightarrow C_1,\ f_{13} \rightarrow C_1,\ f_{23} \rightarrow C_3$; and the Final Class is *Water*
(b) $f_{12} \rightarrow C_2,\ f_{13} \rightarrow C_1,\ f_{23} \rightarrow C_2$; and the Final Class is *Vegetation*
(c) $f_{12} \rightarrow C_1,\ f_{13} \rightarrow C_3,\ f_{23} \rightarrow C_3$; and the Final Class is *Built-up*
(d) $f_{12} \rightarrow C_1,\ f_{13} \rightarrow C_1,\ f_{23} \rightarrow C_2$; and the Final Class is *Water*

[3 *Mark*]

Correct answer: (a) $f_{12} \rightarrow C_1,\ f_{13} \rightarrow C_1,\ f_{23} \rightarrow C_3$; and the Final Class is *Water*

Detailed Solution:

The total number of classifiers is $c(c - 1)/2 = 3$.

Given that the new unclassified pixel has spectral features: (x_1, x_2) = (2.5,1.2), i.e., the feature values are: x_1=2.5 and x_2 = 1.2.

Evaluating each binary classifier:

Substituting the feature values x_1=2.5 and x_2 =1.2 into the equations for each trained classifier:

Evaluating C_1 vs C_2:

$$f_{12}(\mathrm{x}) = \mathrm{x}_1 - \mathrm{x}_2 - 0.5 = 2.5 - 1.2 - 0.5 = 0.8$$

Evaluating C_1 vs C_3:

$$f_{13}(\mathrm{x}) = \mathrm{x}_1 + \mathrm{x}_2 - 3 = 2.5 + 1.2 - 3 = 0.7$$

Evaluating C_2 vs C_3:

$$f_{23}(\mathrm{x}) = 2\mathrm{x}_2 - \mathrm{x}_1 - 1 = 2(1.2) - 2.5 - 1 = -1.1$$

Determining the votes from each classifier:

Classifier Pair	**Decision function $f_{ij}(x)$**	**Rule (vote)**	**Vote** Cast

C_1 vs C_2	$f_{12}(x_1, x_2) = 0.8$	If $f_{12} > 0$, vote C_1; else vote C_2	C_1
C_1 vs C_3	$f_{13}(x_1, x_2) = 0.7$	If $f_{13} > 0$, vote C_1; else vote C_3	C_1
C_2 vs C_3	$f_{23}(x_1, x_2) = -1.1$	If $f_{23} > 0$, vote C_2; else vote C_3	C_3

Classification of new pixel:

As per the votes casted, we can see that, the received votes are: $C_1 = 2, C_2 = 0 \; and \; C_3 = 1$.

As the highest no. of votes were received by C_1, so, the One-vs-One SVM classifies the point as Class C_1 (Water).